Hamid Osooli

University of Illinois Urbana-Champaign (UIUC), Champaign, Il

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SKILLS

Languages: Python (expert), C/C++ (intermediate), MATLAB (expert), Java (familiar), LaTeX(expert)
ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, TRL, PEFT, DeepSpeed, Accelerate, scikit-learn
MLOps: Docker, AIchor, Weights & Biases, Neptune, Git
Databases: MySQL, SQLite, MongoDB
Robotics: ROS/ROS 2, Robotic Eye (author and maintainer), Duckiebot (DB-J), Turtlebot 3 & 4, Sensor fusion
Simulation Tools: Genesis, Gazebo, Simulink, Adams, Ansys, 20-sim

EXPERIENCE

PhD Candidate in Aerospace Engineering **08 2024 – present**

- **Robotics**
- Designed a grid following behavior (fixed trajectory and real-time individual policies) for the turtlebot 3 robots using PID controller for the odometry, and Vicon/Qualysis motion capture systems in ROS 2, *(Ubuntu 22, ROS 2 Humble)*
- Designed a ROS 2 publisher node for the Qualysis motion capture system
- Extended the grid following behavior to multiple robots, through parallel commands in python over different ROS_DOMAIN_IDs, and ROS_NAMESPACE
- Implemented Independent Q-Learning (IQL) on physical TurtleBot 3 robots for real-world multi-robot experiments using the Qualisys motion capture system for robot localization and tracking. *Course Project for Formal Methods in AE Robotics*
- Implemented forward/inverse kinematics, RRT-Connect motion planning with OMPL, path smoothing, and collision checking for robotic manipulators; Developed trajectory optimization and model predictive control (MPC) for planar drones and a UR5e manipulator with obstacle avoidance; implemented PPO reinforcement learning and behavioral cloning using MPC expert demonstrations; Trained and evaluated PPO-based locomotion policies for a Unitree Go1 quadruped in MuJoCo Playground to learn animal-inspired locomotion policies, including rabbit- and horse-like gait patterns. *Course Projects for Advanced Robotic Planning*
- **Multi-Agent Reinforcement Leaning**
- Created wrapper functions for the `gym multigrid` environment within the Epymarl library
- Implemented difference rewards multi-agent actor critic, and learnable multi-agent actor critic credit assignment algorithms using attention mechanism based on MAPPO within the Epymarl library
- Implemented Counterfactual Policy Gradient (COMA) paper from scratch

AE 483-Autonomous Systems Lab **08 2024/2025/2026 – 12 2024/2025/2026**

- Teaching Assistant, Instructor: Prof. Timothy Bretl
- Transformed 3D tracking data using the Qualysis motion capture system for drone applications.
- Calculated inertial parameters to enhance drone performance and stability.
- Designed and tuned a Linear Quadratic Regulator (LQR) controller for optimized drone control.
- Improved drone accuracy through Root Mean Square Error (RMSE) analysis.
- Implemented both offline and online observers for real-time drone monitoring and control.

Persistent Autonomy and Robot Learning Lab (PeARL) **09 2021 – 08 2024**

- Research Assistant, Advisor: Dr. Reza Azadeh (Ahmadzadeh)
- **Robotics**
- Designed, conducted, and recorded the real-world experiments on ROS 2 with Turtlebot 4 *(Ubuntu 20, ROS 2 Galactic)*, paper published at MRS 2023
- Calibrated the onboard camera and implemented a PID-based lane-following controller on a Duckiebot, enabling autonomous navigation of an oval track using vision-based yellow-lane detection; achieved 14+ continuous laps, exceeding the 5-lap requirement for full marks and earning an A+. *Course Project for Fundamentals of Robotics*
- **Multi Agent Reinforcement Leaning**
- Developed a multi agent search and rescue environment with Independent Q Learning (IQL) algorithm
- Conducted experiments on Multi-Agent Deep Deterministic Policy Gradient (MADDPG)
- Devised a game theory hierarchical algorithm and presented the work at the Northeast Robotics Colloquium (NERC 2022)
- Implemented the Distributed, and Independent Q-learning algorithms
- **Reinforcement Leaning**
- Implemented the fundamental RL algorithms from Sutton & Barto's textbook entirely from scratch
- **Other**

- Implemented the *Attention is All You Need* paper for the Deep Learning for NLP course project.
- Supervised two high school students in creating a GitHub repository for a game theory algorithm designed to handle occlusion with the robotic eye, article published at Special Issue Sensors Journal: Feature Papers in Intelligent Sensors 2024
- Developed a multi robot task assignment framework, paper published at IROS 2023 Advances in Multi-Agent Learning - Coordination, Perception, and Control Workshop
- Performed experiments on agent's competency in belief space

New England Robotics Validation and Experimentation (NERVE) Center **09 2023 – 08 2024**

- Research Assistant, Advisor: Dr. Reza Azadeh (Ahmadzadeh)
- **Robotics**
- Designed and 3D printed a pressure sensor holder for the exoskeleton by Autodesk Inventor
- **Reinforcement Leaning**
- Implemented Dyna-style reinforcement learning on MyoPro exoskeleton hardware for real-time adaptation to user interactions, integrating an Unscented Kalman Filter (UKF), Locally Weighted Projection Regression (LWPR), and APDM IMU sensor data.
- **Machine Learning**
- Modeling the Myopro exoskeleton using machine learning *(Locally Weighted Projection Regression (LWPR), K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Gaussian Process Regression (GPR), eXtreme Gradient Boosting (XGBoost), Multi Layer Perceptron (MLP))* by fusing the data from accelerometers, gyroscopes and magnetometers in APDM IMU sensors, papers published at UR 2024 (award finalist)

Game Theory for UAV Cinematography **09 2018 – 07 2019**

- Devised an integrated control algorithm based on game theory
- Developed MATLAB simulation for car tracking with UAV
- Created a simulation for UGV tracking using a game theory algorithm with a camera-mounted UAV *(Ubuntu 18, ROS Melodic, and PX4 10.1)*

Making a Robot Inspired by the Human Eye **09 2015 – 06 2016**

- Designed a robotic eye, inspired by the human eye, from scratch (CAD model, PID controller, closing the loop from visual input to the servomotors) papers published at the ISME 2017 and UR 2023
- Developed the XL_320 MATLAB toolbox for robot servomotors
- Calibrated the cameras for stereo vision
- Devised game theory control algorithms for the robotic eye, paper published at the ICEE 2019

EMPLOYMENT

InstaDeep **06 2025 – 12 2025**

Research Engineer Intern *San Francisco, CA*

- Conducted research on improving weak-to-strong safety alignment (helpfulness and harmlessness) in large language models (LLMs) using reinforcement learning techniques.
- Designed reward models and implemented Reinforcement Learning from Human Feedback (RLHF) with Proximal Policy Optimization (PPO) on Llama 3.1 8B and Llama 3.2 3B Instruct models.
- Analyzed bias–variance trade-offs and developed deception score metrics to evaluate weak-to-strong alignment performance.
- Built and managed containerized LLM training and evaluation pipelines using AIchor, Docker, DeepSpeed, and Hugging Face Accelerate, deploying distributed RLHF workloads on NVIDIA H100 GPU clusters through Git-based experiment workflows.

PANTOhealth **07 2020 – 08 2021**

Data Scientist *Berlin, Germany*

- Optimized simulation algorithms for rail vehicle pantograph-catenary interactions, reducing runtime from 129 seconds to 39 seconds.
- Enhanced computational performance by developing a state-space model simulation in MATLAB and Python.
- Customized the `scipy.integrate.solve_ivp` back-end for efficient ODE solutions under specific conditions.

EDUCATION

University of Illinois Urbana-Champaign (UIUC) **08 2024 – present**

PhD Candidate in Aerospace Engineering (Robotics and AI), GPA: 3.86/4,

University of Massachusetts Lowell (UML) **09 2021 – 08 2024**

Thesis: A Multi-Robot Task Assignment Framework for Search and Rescue with Heterogeneous Teams

MSc in Computer Science (Robotics and AI), GPA: 3.78/4,

SELECTED PUBLICATIONS

Evaluating Risks in Weak-to-Strong Alignment: A Bias-Variance Perspective
Association for the Advancement of Artificial Intelligence (Under Review at AAAI 2027)
Hamid Osooli, Kareema Batool, Rick Gentry, Tiasa Singha Roy, Ashwin Gupta, Anirudha Ramesh

Deeper Dive Into Multi-Agent Credit Assignment
Preprint (Under Revision)
Niket Parikh*, Hamid Osooli*, Huy T. Tran

Zero-Shot Coordination in Ad Hoc Teams with Generalized Policy Improvement and Difference Rewards
25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)
Rupal Nigam, Niket Parikh, Hamid Osooli, Mikihisa Yuasa, Jacob Heglund, Huy T. Tran

S2Act: Simple Spiking Actor
Preprint (Under Revision)
Ugur Akcal, Seung Hyun Kim, Mikihisa Yuasa, Hamid Osooli, Jiarui Sun, Ribhav Sahu, Mattia Gazzola, Huy T. Tran, Girish Chowdhary

Coordination Machines for Minimizing Communication in Multi-Agent Reinforcement Learning
Preprint (Under Revision)
Jacob Heglund, Mikihisa Yuasa, Hamid Osooli, Huy T. Tran

Influence of Team Interactions on Multi-Robot Cooperation: A Relational Network Perspective
International Symposium on Multi-Robot & Multi-Agent Systems (MRS 2023)
Yasin Findik*, Hamid Osooli*, Paul Robinette, Kshitij Jerath, S. Reza Ahmadzadeh

A Multi-Robot Task Assignment Framework for Search and Rescue with Heterogeneous Teams
The 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023 Multi-Agent Learning Workshop)
Hamid Osooli, Paul Robinette, Kshitij Jerath, S. Reza Ahmadzadeh

COURSEWORK

- Robotics
- Fundamentals of Robotics
- Human Robot Interaction
- Advanced Robotic Planning
- Formal Methods in AE Robotics
- Algorithms
- Reinforcement Learning
- Statistical Reinforcement Learning
- Deep Learning
- Deep Learning for NLP
- Evaluation of LLMs
- Advanced Control
- Nonlinear Control
- Optimal Control
- Automatic Control